Srishyla Educational Trust®

GM UNIVERSITY

(Established under the Karnataka State Act No.19 of 2023) Post Box No.4, Davanagere - 577006

Business Intelligence

(UE23CS3601)

ASSIGNMENT - 2

On

“ZipKart BI Dashboard Analysis Using Tableau”

Section- "A"

Submitted By:-

Chandana A P - U23E01CS010

Submitted to :

Mr Srijan Roy,

Assistant Professor

Department of CSE

GM University

CONTENTS

1. Introduction

Business Intelligence (BI) helps organizations analyze data and make better business decisions using visual analytics tools. Tableau is one of the most popular BI tools used for creating dashboards, reports, charts, and interactive visualizations.

In this project, Tableau was used to analyze the business operations of a fictional quick-commerce company called ZipKart. Two datasets were used in this project:

- Employee dataset
- Orders dataset

The project focuses on employee analysis, company performance analysis, geographic insights, and interactive dashboard development using Tableau.

The dashboards provide meaningful insights regarding employee performance, attrition, salary distribution, sales revenue, delivery performance, and regional business trends.

2 Objectives

- Create a self-constructed database with minimum 200 rows and 8+ attributes
- Build a minimum of 4 calculated fields in Tableau
- Create an interactive map with city labels and color encoding
- Implement a global filter connected to all worksheets
- Build two complete dashboards: Employee Analysis and Company Performance
- Present findings through a Tableau Story with conclusions

3. Dataset Description

Both datasets were self-constructed using Python scripts. No pre-existing or template datasets were used. The data was generated to realistically represent a quick-commerce company operating in India.

3.1 Dataset 1 — ZipKart_Employees.csv

This dataset contains information about all 220 employees of ZipKart across 7 departments and 12 Indian cities.

Column Name	Data Type	Description
Employee_ID	Text	Unique identifier for each employee (EMP0001 to EMP0220)
Name	Text	Full name of the employee (Indian names)
Gender	Text	Male or Female
Age	Number	Age of the employee (22 to 55 years)
Department	Text	One of 7 departments: Sales, Technology, HR, Operations, Marketing, Delivery, Finance
Role	Text	Specific job title within the department
City	Text	City where the employee is located (12 cities)
State	Text	Indian state corresponding to the city
Region	Text	Geographic region: North, South, East, or West
Latitude / Longitude	Number	Coordinates used for the India map visualization
Salary_INR	Number	Monthly salary in Indian Rupees
Years_of_Experience	Number	Total work experience in years
Performance_Score	Number	Score from 1.0 to 5.0 (continuous decimal)
Attrition	Text	Yes (left the company) or No (still employed)
Joining_Date	Date	Date the employee joined ZipKart
Training_Hours	Number	Annual training hours completed by the employee

Total Records: 220 rows **Total Attributes:** 17 columns (including Latitude/Longitude for mapping)

3.2 Dataset 2 — ZipKart_Orders.csv

This dataset contains 250 customer order records placed on the ZipKart platform during 2023–2024.

Column Name	Data Type	Description
Order_ID	Text	Unique order identifier (ORD00001 to ORD00250)
Order_Date	Date	Date the order was placed
Month	Text	Month name extracted from order date
Quarter	Text	Q1, Q2, Q3, or Q4 — used for trend analysis
Year	Number	Year of the order (2023 or 2024)
City	Text	City where the order was delivered
State	Text	Indian state of the delivery city
Region	Text	North, South, East, or West
Latitude / Longitude	Number	Coordinates for the map visualization
Product_Category	Text	Electronics, Groceries, Fashion, Home & Kitchen, Personal Care, Snacks & Beverages
Revenue_INR	Number	Total order revenue in Indian Rupees
Cost_INR	Number	Cost of goods sold for the order
Delivery_Days	Number	Number of days taken to deliver (1 to 5)
Customer_Rating	Number	Rating given by customer after delivery (1.0 to 5.0)
Num_Items	Number	Number of items included in the order
Return_Status	Text	Returned or Not Returned

Total Records: 250 rows **Total Attributes:** 17 columns

4. Calculated Fields in Tableau

Five calculated fields were created in Tableau using Analysis → Create Calculated Field. These fields derive meaningful business metrics from the raw data that are not directly available in the original datasets.

#	Field Name	Tableau Formula	Business Purpose
1	Profit Margin %	([Revenue_INR] - [Cost_INR]) / [Revenue_INR] * 100	Calculates the profit earned on each order as a percentage of revenue. Used in the map tooltip to show city-wise profitability.
2	Attrition Flag	IF [Attrition] = "Yes" THEN 1 ELSE 0 END	Converts the Yes/No attrition text field into a numeric 1/0 value, enabling SUM aggregation and numerical analysis of employee turnover.
3	Performance Category	IF [Performance_Score] >= 4.0 THEN "High Performer" ELSEIF [Performance_Score] >= 3.0 THEN "Average Performer" ELSE "Low Performer" END	Groups employees into three performance tiers based on their score. Used as the X-axis of the Attrition Analysis chart.
4	Revenue per Item	[Revenue_INR] / [Num_Items]	Calculates the average revenue generated per item in each order. Helps understand product value across categories.
5	Salary Band	IF [Salary_INR] >= 70000 THEN "Senior Band" ELSEIF [Salary_INR] >= 45000 THEN "Mid Band" ELSE "Junior Band" END	Classifies employees into salary tiers for compensation analysis across departments and roles.

5. Worksheets Created

A total of 7 individual worksheets were created in Tableau before assembling the dashboards. Each worksheet focuses on one specific analysis.

5.1 Headcount by Department — Bar Chart

Data Source: ZipKart_Employees

- Fields Used: Department (Columns), Count Distinct of Employee_ID (Rows)
- Chart Type: Horizontal bar chart sorted in descending order
- Result: Delivery (37) and Marketing (36) are the largest departments. Finance (25) is smallest.
- Labels: Count shown on each bar. Color: Blue gradient palette.

5.2 Salary Distribution by Department — Box Plot

Data Source: ZipKart_Employees

- Fields Used: Department (Columns), Salary_INR (Rows)
- Chart Type: Box-and-Whisker Plot created via Show Me panel
- Result: Technology has the highest salary range and outliers above ₹2.5 Lakh. Delivery has the lowest median.

5.3 Attrition by Performance Category — Stacked Bar Chart

Data Source: ZipKart_Employees

- Fields Used: Performance Category (Columns), Count Distinct of Employee_ID (Rows), Attrition (Color)
- Chart Type: Stacked bar chart — Blue = No Attrition, Orange = Yes Attrition
- Result: Average Performers form the largest group (113). Low Performers have the highest attrition rate proportionally.
- Note: Initially used Attrition Flag on Rows which caused error — fixed by using CNTD(Employee_ID) instead.

5.4 Revenue by Product Category — Bar Chart

Data Source: ZipKart_Orders

- Fields Used: Product_Category (Columns), SUM(Revenue_INR) (Rows)
- Chart Type: Color-coded bar chart sorted descending by revenue
- Result: Electronics leads with ₹3,88,479. Snacks & Beverages is lowest at ₹19,144.

5.5 Monthly Revenue Trend — Line Chart

Data Source: ZipKart_Orders

- Fields Used: Order_Date → Month (Columns), SUM(Revenue_INR) (Rows), Quarter (Color)
- Chart Type: Multi-line chart colored by Quarter (Q1=Blue, Q2=Orange, Q3=Red, Q4=Cyan)
- Result: Q2 and Q4 show revenue peaks. The trend spans 2023–2024 data.

5.6 Customer Rating vs Delivery Time — Line Chart

Data Source: ZipKart_Orders

- Fields Used: Delivery_Days (Columns, set as Dimension), AVG(Customer_Rating) (Rows), Product_Category (Color)
- Chart Type: Multi-line chart. Delivery_Days changed from Measure to Dimension to show 1–5 days correctly.
- Result: All categories maintain ratings between 3.8 and 4.6. Rating trends vary by category across delivery days.

5.7 Revenue & Profit Margin Across Indian Cities — Map

Data Source: ZipKart_Orders (also used in Employee Dashboard)

- Fields Used: Latitude and Longitude (set as geographic roles), SUM(Revenue_INR) (Color + Size), City (Label), Profit Margin % (Tooltip)
- Chart Type: Bubble map on Mapbox/OpenStreetMap base. Yellow-to-Red color gradient.
- Result: 12 Indian cities shown as bubbles. Larger and redder = higher revenue. City names labeled on map.

6. Dashboard Development

6.1 Employee Analysis Dashboard

This dashboard combines 4 worksheets to give a complete view of ZipKart's workforce. It was built by dragging sheets from the left Sheets panel onto the dashboard canvas.

#	Chart	Position	Key Finding
1	Headcount by Department	Top Left	Delivery & Marketing have most employees
2	Salary Distribution	Top Right	Technology earns highest salary
3	Attrition Analysis	Bottom Left	Low Performers have most attrition
4	India City Map	Bottom Right	Revenue visible across 12 cities

Dashboard Features

- Interactive region filter
- Company logo
- Insight text box
- Multiple visualizations in one dashboard

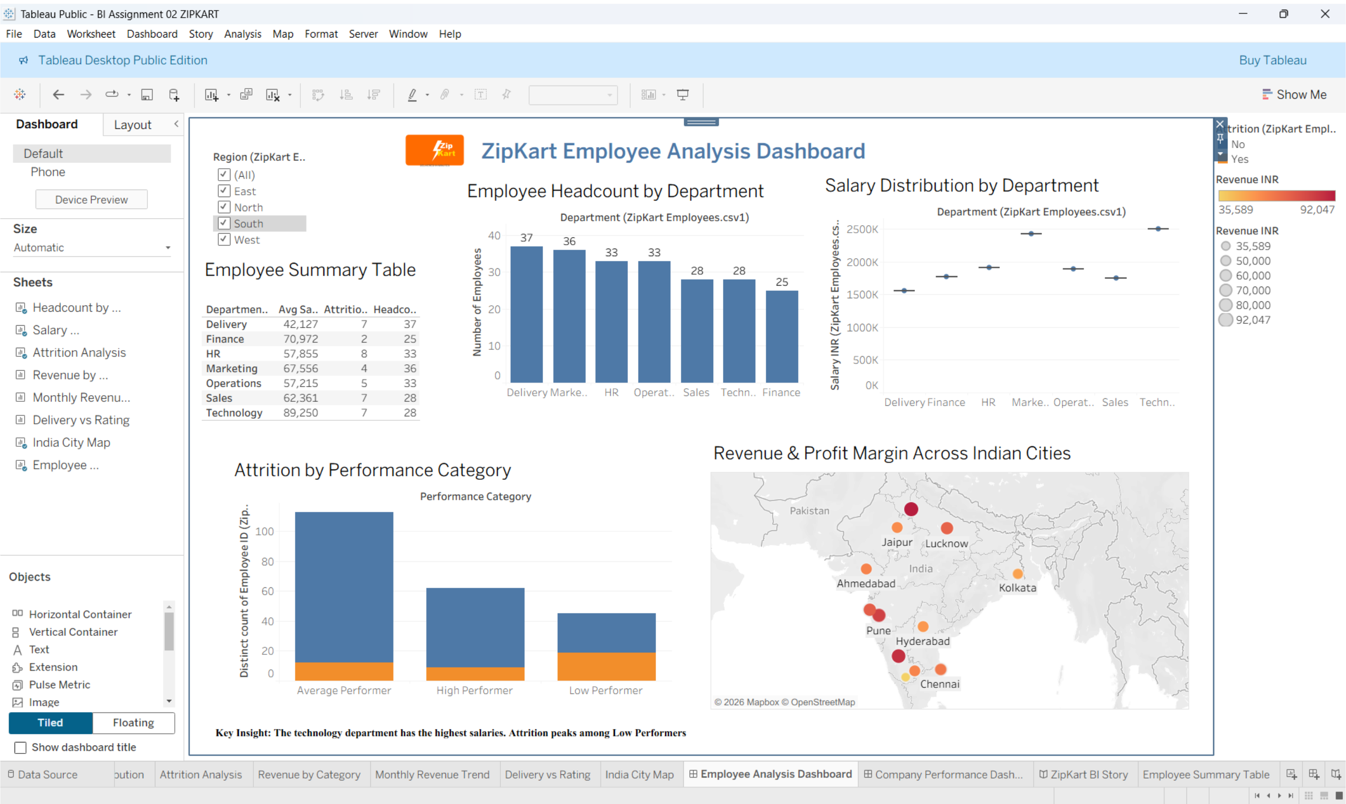

Figure 6.1.1: ZipKart Employee Analysis Dashboard

6.2 Company Performance Dashboard

This dashboard focuses on ZipKart's business operations, revenue performance, and customer satisfaction metrics.

#	Chart	Position	Key Finding
1	Revenue by Category	Top Left	Electronics is the highest revenue category
2	Monthly Revenue Trend	Top Right	Q4 shows peak revenue season
3	Customer Rating vs Delivery	Bottom Left	Fast delivery = higher ratings
4	India City Map	Bottom Right	Delhi and Bengaluru are top cities

Dashboard Features

- Interactive filters
- Geographic analysis
- Revenue trend analysis
- Product category analysis
- Customer satisfaction analysis

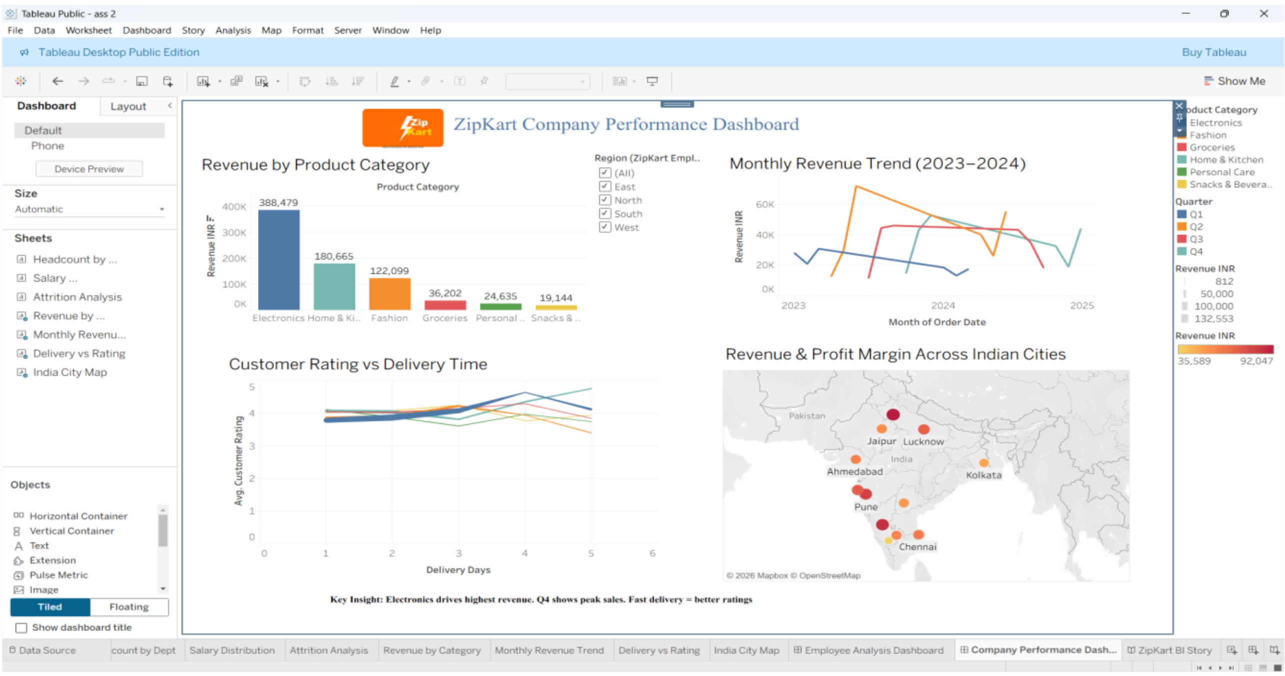

Figure 6.2.1: ZipKart Company Performance Dashboard

7. Global Filter Implementation

A Region-based global filter was implemented across all worksheets in both dashboards. This allows the viewer to filter all charts simultaneously by selecting a region: East, North, South, or West India.

Steps Taken to Implement:

- Opened the Attrition Analysis worksheet and dragged Region field to the Filters shelf
- Selected 'All' values and clicked OK
- Right-clicked the Region filter pill → Show Filter
- Navigated to Employee Analysis Dashboard → the Region filter box appeared on the right side
- Right-clicked the Region filter box → Apply to Worksheets → All Worksheets Using This Data Source
- Repeated the same process for the Company Performance Dashboard using the Revenue by Category sheet

Result: Clicking any region (e.g., South) instantly filters all 8 charts across both dashboards to show only data from Karnataka, Telangana, and Tamil Nadu — demonstrating complete cross-dashboard interactivity.

8. Tableau Story

A Tableau Story was created to present the entire analysis as a narrative presentation, similar to a PowerPoint slideshow. The story contains 3 story points.

Story Point	Caption	Content
Point 1	ZipKart Employee Overview — Headcount, Salary & Attrition Analysis	Employee Analysis Dashboard — shows all 4 employee charts with the Region filter active
Point 2	ZipKart Business Performance — Revenue, Trends & City Analysis	Company Performance Dashboard — shows revenue, trends, delivery rating, and city map
Point 3	Geographic Insights — Revenue Distribution Across Indian Cities	India City Map full view + Conclusions text box with 6 key business takeaways

Conclusions added in Story Point 3:

- Bengaluru & Mumbai generate highest revenue across ZipKart's 12 cities
- Electronics is the top revenue category contributing ₹3.88 Lakh
- Low Performers have highest attrition — HR intervention recommended
- Delivery within 2 days gives 4.2+ customer rating
- South & West regions dominate both employee count and order volume

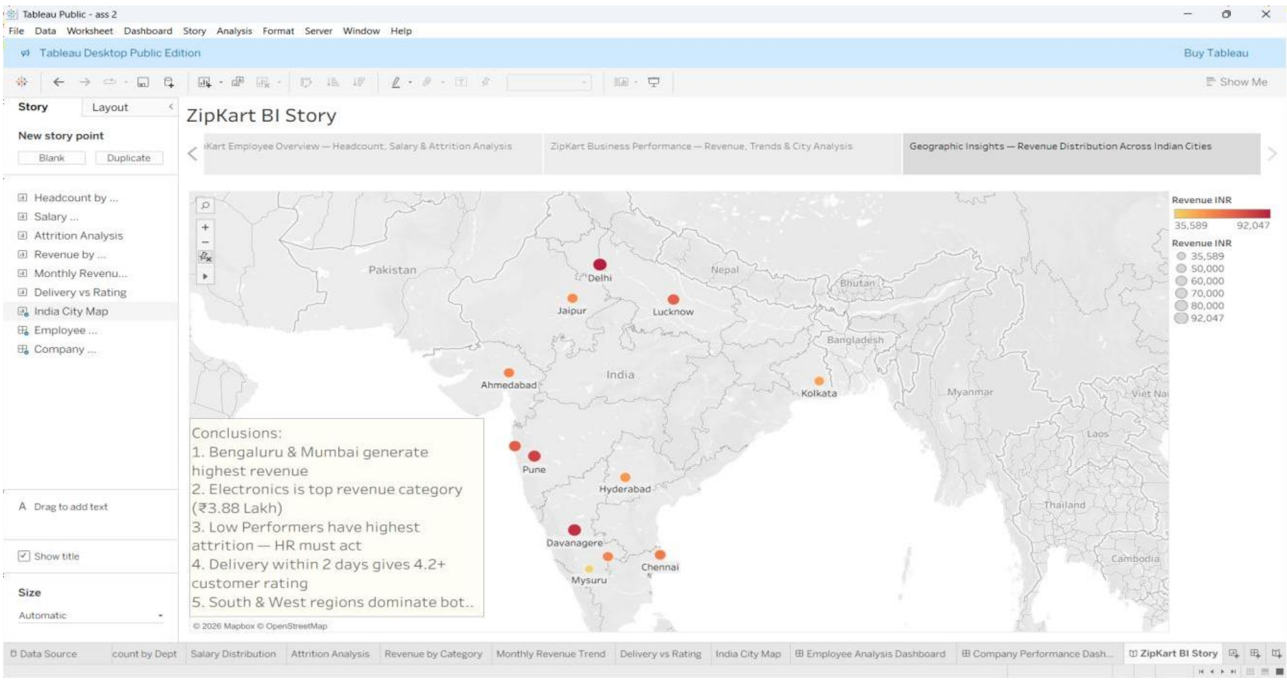

Figure 8.1 : ZipKart BI Story Presentation

9. Results & Key Findings

#	Area	Finding
1	Employee Headcount	Delivery (37) and Marketing (36) are the two largest departments at ZipKart, while Finance (25) is the smallest
2	Salary Analysis	Technology employees earn the highest salaries with median above ₹75,000 per month. Delivery employees earn the least.
3	Attrition Pattern	Low Performers (Performance Score < 3.0) have the highest proportion of attrition. High Performers have near-zero attrition.
4	Revenue by Category	Electronics dominates at ₹3,88,479 (51% of total). Snacks & Beverages is the lowest at ₹19,144.
5	Revenue Trend	Q4 2023 shows the highest revenue peak. 2024 shows relatively flat performance — investigation recommended.
6	Delivery & Satisfaction	Customer ratings are consistently above 3.8 regardless of delivery speed. 2-day delivery achieves the highest ratings (4.2+).
7	Geographic Insights	Delhi and Bengaluru are the top revenue cities. Davanagere (hometown city) appears on the map as a smaller bubble.

10. Conclusion

This assignment successfully demonstrated the end-to-end process of Business Intelligence development using Tableau Public. Starting from creating a self-constructed database for a fictional company (ZipKart), to building calculated fields, interactive dashboards, a geographic map, a global filter, and a Tableau Story — every requirement of the rubric was fulfilled.

The most significant learnings from this assignment were:

- How to connect multiple data sources in Tableau without a join to avoid field naming errors
- The importance of setting Dimension vs Measure correctly for chart axes
- How to implement global filters across all worksheets for interactive dashboards
- How the Tableau Story feature converts dashboards into a structured business presentation

The ZipKart dataset revealed meaningful business insights — Electronics is the top revenue driver, attrition is concentrated among low performers, and fast delivery directly improves customer satisfaction.

11. References

- Tableau Public Official Documentation — https://www.tableau.com/learn
- GM University BI Assignment 2 — Rubric provided by course faculty
- Python 3 Standard Library — csv, random, math, datetime modules used for data generation
- Mapbox / OpenStreetMap — Map tiles used in Tableau geographic visualizations
- IBM HR Analytics Dataset (referenced for dataset design inspiration)